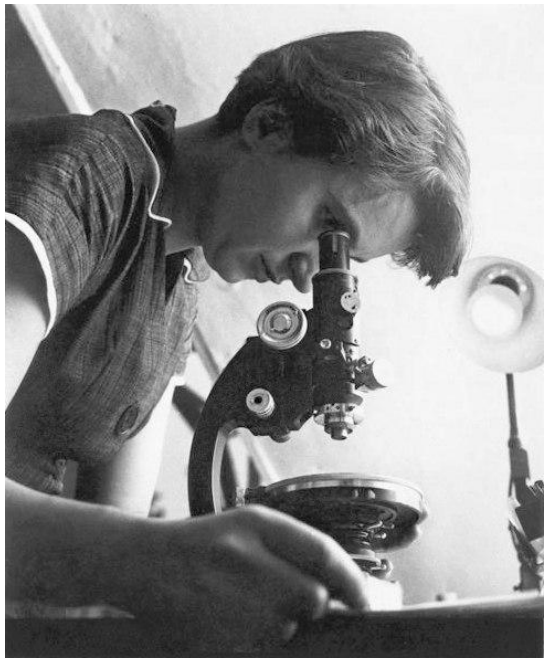
CY4001: Industrial Chemistry

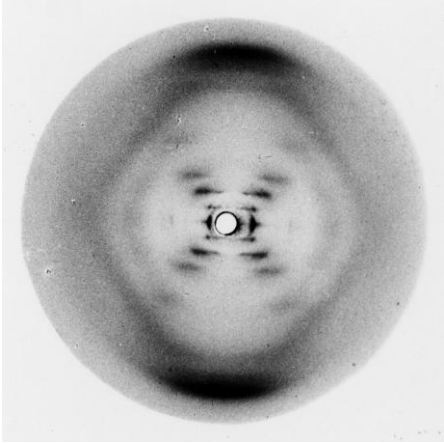
Unit 3:Fuel Chemistry
(Solid Fuels)

Rosalind Franklin



- Used X-ray diffraction to study coal's internal carbon structure with unprecedented precision.
- Discovered the difference between graphitizing and non-graphitizing carbons
- Explained how coal's microstructure changes on heating, influencing how efficiently it burns.
- Her research clarified why some coals form strong coke for steelmaking while others crumble.

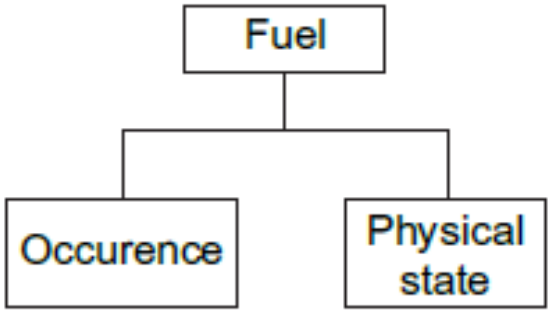
- Took Photo 51, the key image revealing DNA's double helix. Her data was shown to James Watson and Francis Crick without permission. Missed the Nobel Prize awarded to Maurice Wilkins, Watson, and Crick, though her work was central to the discovery.



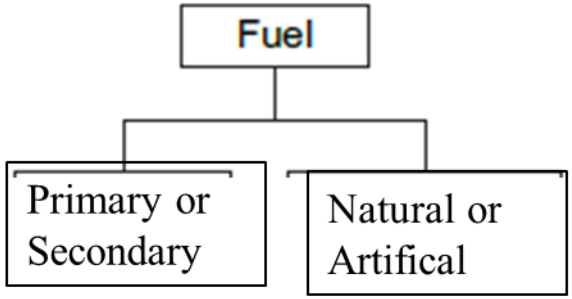
Let your work shine so brightly that the world cannot ignore it

Unit 3: Fuel Chemistry

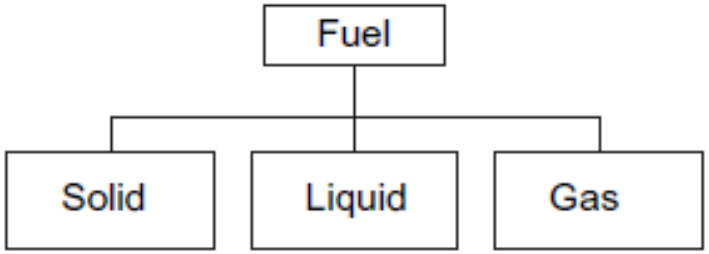
Classification of Fuels



On the basis of occurrence



On the basis of physical state



Physical state	*Primary fuel*	*Secondary fuel*
Solid	Wood, Peat, Coal, Lignite	Coke, Charcoal
Liquid	Crude petroleum	Petrol, Diesel, Synthetic
Gas	Natural gas	Producer gas, Water gas, Coal gas, Biogas, LPG

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Solid Fuels

- Solid fuels are classified into natural fuels, and manufactured fuels.

Natural fuels	*Manufactured fuels*
	Solid fuels
Wood	Tanbark, Bagasse, Straw
Coal	Charcoal
Oil shale	Coke
	Briquettes

Advantages of Solid Fuels

- They are easy to transport.
- They are convenient to store without any risk of spontaneous explosion.
- Their cost of production is low.
- They possess moderate ignition temperature.

Disadvantages of solid Fuels

- Their ash content is high.
- Their large proportion of heat is wasted.
- They burn with clinker formation.
- Their combustion operation cannot be controlled easily.
- Their cost of handling is high.

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Woods and their Characteristics

- **Wood is a vegetable tissue of trees and bushes. It consists of mainly cellular tissue and lignin and lesser parts of fat and tar, as well as sugar.**

The main constituents (%) of several kinds of wood

Type of wood	Water	Sugar	Fat-tar	Cellular tissue	Lignin
Beach wood	12.57	2.41	0.41	45.57	39.14
Birch wood	12.48	2.65	1.14	55.62	28.21
Fir (Boot)	13.87	1.26	0.97	55.90	26.91
Pine wood	12.87	4.05	1.63	53.27	28.18

Cellular Tissue and Lignin (in %)

Constituents	Cellular tissue	Lignin
Carbon	44.4	54.58
Hydrogen	6.2	5.8–6.3
Oxygen	49.4	35.39

Calorific values of wood

Kinds of wood	Lowest calorific value (cal/kg)	Highest calorific value (cal/kg)
Oak	4729	4750
Birch	4695	4831
Elm	4674	4833
Alder	4745	4839
Pine	4818	5310
Fir	4887	4900
Lrch	4775	4840

Average values of constituents of wood

Element	Pine wood	Birch wood	Oak wood
C	50.05	48.45	49.8
H	6.04	5.95	5.81
O + N	43.21	45.26	44.00
Ash	0.70	0.34	0.4

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Coals and Their Characteristics

- Coal is a mineral substance of vegetable origin and a hard rock which can be burned as a solid fossil fuel.

- The large deposits of coal in India are in Bengal, Bihar and Madhya Pradesh. Most of the Indian coal is of low grade variety and coal washing to obtain low ash metallurgical coal is inevitable.

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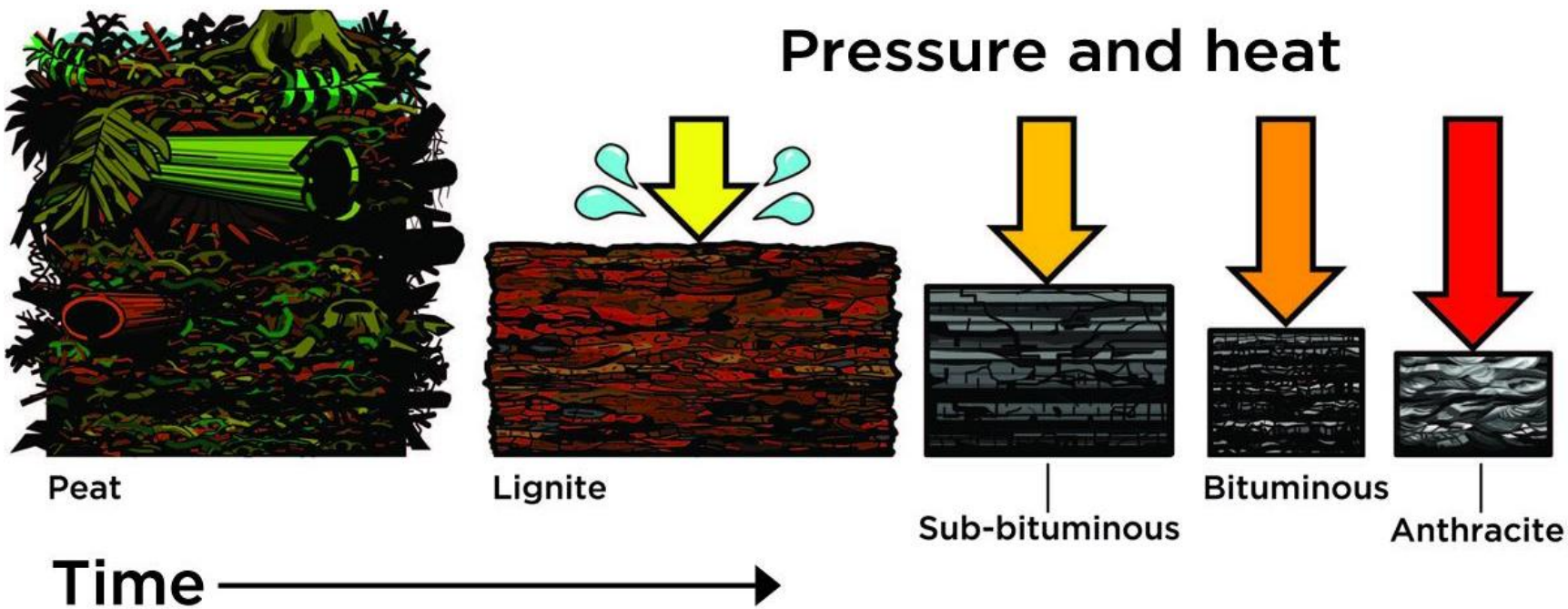
2.1 Ranking of Coal

- Coal is classified on the basis of its rank.
- The rank of coal denotes its degree of maturity.
- Vegetable matter, under the action of pressure, heat and anaerobic conditions, gets converted into different stages of coal, namely,

Peat → Lignite → Sub-bituminuous → Bituminous → Anthracite



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Peat:

- Peat is the surface organic layer of a soil that consists of partially decomposed organic matter, derived mostly from plant material.

- Peat, considered to be a precursor of coal, has industrial importance as a fuel in many regions, for example, Ireland and Finland.

- In its dehydrated form, peat is a highly effective absorbent for fuel and oil spills on land and water.

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Lignite:

- Lignite, often referred to as brown coal, is a soft, brown, combustible, sedimentary rock formed from naturally compressed peat.

- Lignite is the youngest coal from the geological perspective. It is a soft coal composed mainly of volatile matter and moisture content with low fixed carbon (refers to carbon in its free state, not combined with other elements).

- Lignite, is the lowest rank of coal and used almost exclusively as fuel for electric power generation.

- Jet is a compact form of lignite that is sometimes polished and has been used as an ornamental stone since the Iron Age.

2.1.3 Sub-bituminous Coal:

- Sub-bituminous coal is a lower grade of coal that contains 35–45% carbon.
- Whose properties range from those of lignite to those of bituminous coal.
- Used primarily as fuel for steam electric power generation.
- It is an important source of light aromatic hydrocarbons for the chemical synthesis industry.

2.1.4 Bituminous Coal:

- Bituminous coal, a dense mineral, is black or sometimes dark brown, often with well-defined bands of bright and dull material.
- It is used primarily as fuel in steam-electric power generation, with substantial quantities also used for heat and power applications in manufacturing and to make coke.

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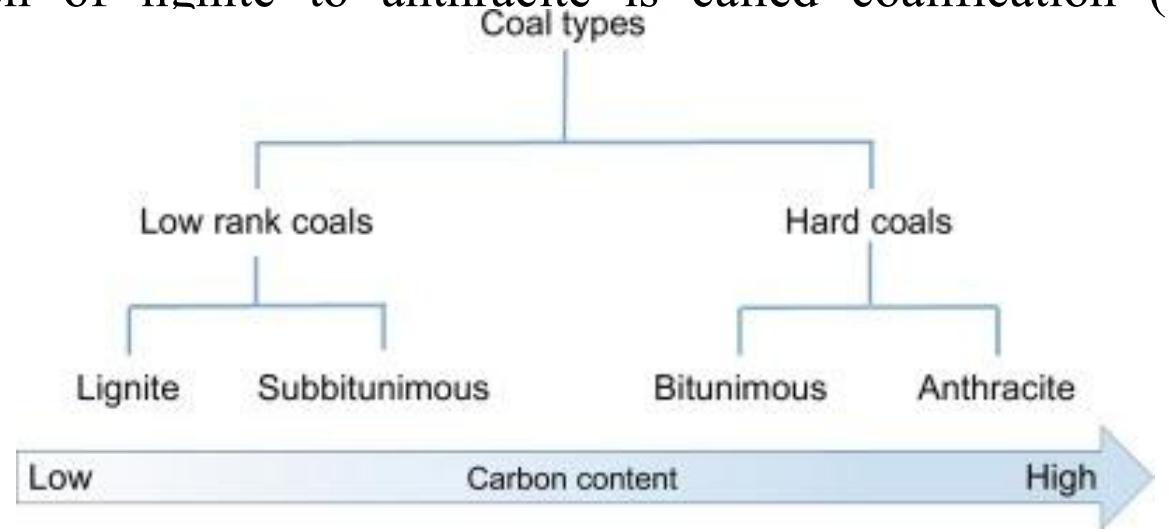
Anthracite

- Anthracite, the highest-ranking coal, is a harder, glossy, black coal used primarily for residential and commercial space heating.

- It may be divided further into metamorphically altered bituminous coal and petrified oil, as from the deposits in Pennsylvania.

- So, peat is the most immature coal, hence it is lowest in rank whereas anthracite is the most matured coal, and hence it is highest in rank.

- The process of conversion of lignite to anthracite is called coalification (or) metamorphism of coal.



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Analysis of Coal

To ascertain the quality and commercial value of coal certain tests are performed to scrutinize its burning properties before it is commercially marketed.

Two commonly used tests are:

- **Proximate analysis**: Used to determines the fixed carbon, volatile matter, moisture and ash percentages.
- **Ultimate analysis**: Used to determine the percentage of carbon, hydrogen, nitrogen and sulphur present in coal.

2.2.1 Proximate analysis

- The determination of the amount of moisture, volatile matter, fixed carbon and ash present in coal is called proximate analysis.

- It is useful in deciding its utilization for a particular industrial use.

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Estimation of Moisture Content in Coal:

- Accurately weigh 1 g of powdered dry coal sample in a previously weighed, clean silica crucible.
- Heat the sample by placing in a hot air oven at 100 to 105°C for one hour. Then the sample is cooled in a desiccator to room temperature and weighed again. The loss in weight represents moisture.

$$\therefore \text{ \% of moisture in coal} = \frac{\text{Loss in weight of coal}}{\text{Weight of coal initially taken}} \times 100$$

Estimation of Volatile Matter (VM) in Coal:

- After the analysis of moisture content, the crucible with the residual coal sample is covered with a lid, and it is heated at 950 ± 20°C for 7.0 minutes in a muffle furnace.
- Percentage of volatile matter can be calculated from the loss in weight of the coal sample.

$$\therefore \text{ \% of volatile matter in coal} = \frac{\text{Loss in weight of moisture-free coal}}{\text{Weight of coal initially taken}} \times 100$$

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Estimation of Ash in Coal:

- After the analysis of volatile matter, the crucible with the residual coal sample is heated without lid at 700 ± 50°C for 30 minutes in a muffle furnace.
- The percentage of ash content can be calculated from the weight of ash formed.

$$\therefore \text{ \% of ash in coal} = \frac{\text{Weight of ash formed}}{\text{Weight of coal taken}} \times 100$$

Estimation of Fixed Carbon:

- Fixed carbon refers to carbon in its free state, not combined with other elements.
- It is determined by subtracting the sum of total moisture, volatile matter and ash contents from 100.

$$\text{\% of fixed carbon} = 100 - \text{\% of [moisture + VM + ash]}$$

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Significance of Proximate Analysis:

- **Moisture:** The lesser the moisture content, the better the quality of coal
- **Volatile Matter:** A good-quality coal should have lesser volatile matter. The coal with higher volatile content ignites easily
- **Ash:** Ash is the combustion product of mineral matters present in the coal. A good-quality coal should contain lesser ash content.
- **Fixed Carbon:** It is the pure carbon present in coal. Higher the fixed carbon content of the coal, higher will be its calorific value.

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Ultimate analysis

- The determination of the weight percentage of carbon, hydrogen, nitrogen, oxygen, and sulfur of pure coal free from moisture and inorganic constituents is called ultimate analysis.

- It is useful in determining the quantity of air required for combustion and the volume and composition of the combustion gases.

- This information is required for the calculation of flame temperature (i.e. temperature attained by the products of combustion) and the flue duct design etc.

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Determination of Carbon and Hydrogen in Coal:

- A known amount of coal is burnt in the presence of oxygen, thereby converting carbon and hydrogen of coal into

$$CO_2 \ (C + O_2 \rightarrow CO_2)$$

$$H_2O \ (H_2 + ½O_2 \rightarrow H_2O)$$

- The products of combustion (H_2O and CO_2) are made to pass over weighed tubes of anhydrous $CaCl_2$ and KOH, which absorb H_2O and CO_2 respectively.

- The increase in the weight of the $CaCl_2$ tube represents the weight of water formed, while increase in the weight of the KOH tube represents the weight of CO_2 formed.

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The percentage of carbon and hydrogen in coal

X — the weight of coal sample taken

Y — the increase in the weight of KOH tube (amount of CO_2)

Z — the increase in the weight of $CaCl_2$ tube (amount of H_2O)

Carbon

$$\underset{12}{C} + \underset{32}{O_2} \longrightarrow \underset{44}{CO_2}$$

44 g of CO_2 contain 12 g of carbon,

i.e, 1 g of CO_2 contains $\frac{12}{44}$ g of carbon.

$\therefore$ Y g of CO_2 contains $\frac{12}{44} \times Y$ g of carbon.

$$\boxed{\% \text{ of C in coal} = \frac{12}{44} \times \frac{Y}{X} \times 100}$$

Hydrogen

$$\underset{2}{H_2} + \underset{16}{1/2\ O_2} \longrightarrow \underset{18}{H_2O}$$

18 g of water contains 2 g of hydrogen.

1 g of water contains 2/18 g of hydrogen.

$\therefore$ Z g of water contains $= \frac{2}{18} \times Z$ g of hydrogen.

$$\boxed{\% \text{ of hydrogen in coal} = \frac{2}{18} \times \frac{Z}{X} \times 100}$$

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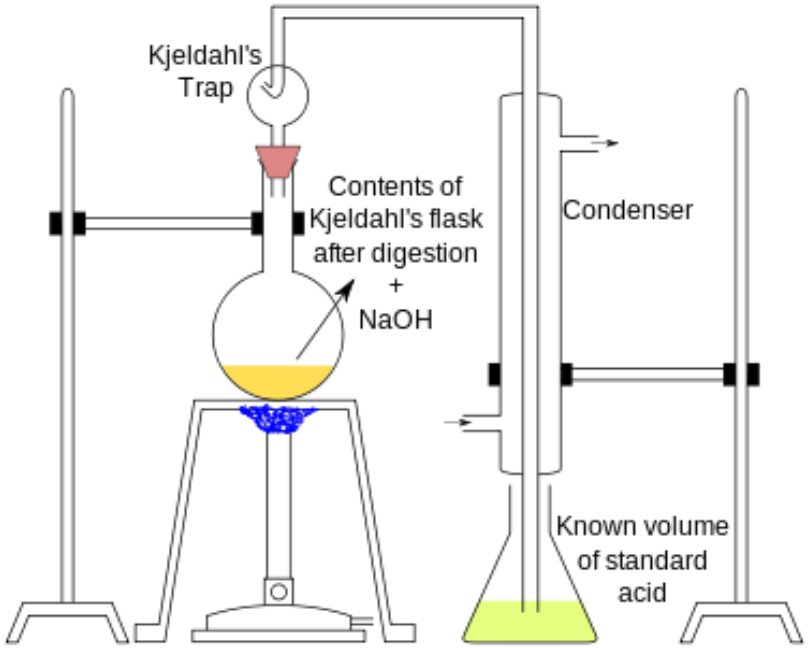
Determination of Nitrogen in Coal (Kjeldahl's method):

- A known amount of powdered coal is heated with conc. H_2SO_4 and K_2SO_4 in a long-necked flask (called Kjeldahl's flask), thereby converting nitrogen of coal to ammonium sulphate.
- When the clear solution is obtained (i.e. the whole nitrogen is converted into ammonium sulphate), it is heated with 50% NaOH solution.

$$(NH_4)_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2NH_3 \uparrow$$

- The ammonia, thus formed, is distilled over and is absorbed in a known quantity of standard 0.1N HCl solution.
- The volume of unused 0.1N HCl is then determined by titrating against a standard NaOH solution.



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The amount of acid neutralized by liberated ammonia from coal is determined.

Let,

Volume of 0.1N HCl initially taken = A ml

Volume of unused HCl = B ml

Acid neutralised by ammonia = $(A - B)$ ml

We know that 1000 ml of 1N HCl = 1 mole of HCl

= 1 mole of NH_3

= 14 g of N_2

$$\therefore (A - B) \text{ ml of } 0.1\text{N HCl} = \frac{14 \times (A - B) \times 0.1}{1000 \times 1\text{N}} \text{ g of } N_2$$

$$X \text{ g of coal sample contains } \frac{14 \times (A - B) \times 0.1}{1000 \times 1\text{N}} \text{ g of } N_2$$

$$\boxed{\% \text{ of nitrogen} = \frac{14 \times \text{Volume of acid consumed} \times \text{Normality}}{1000 \times \text{weight of coal sample } (X)} \times 100}$$

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Determination of Sulphur in Coal:

- A known amount of coal is burnt completely in a bomb calorimeter in the presence of oxygen. Ash, thus obtained, contains sulphur of coal as sulphate, which is extracted with dil HCl.
- The acid extract is then treated with $BaCl_2$ solution to precipitate sulphate as $BaSO_4$.The precipitate is filtered, washed, dried and weighed to calculate the sulphur in coal as stated below:

Let,

X = weight of coal sample taken

M = weight of $BaSO_4$ precipitate formed.

$$\underset{32}{S} + 2O_2 \longrightarrow SO_4 \longrightarrow \underset{233}{BaSO_4}$$

233 g of $BaSO_4$ contains 32 g of sulphur.

i.e. 1 g of $BaSO_4$ contains 32/233 g of sulphur.

$\therefore$ M g of $BaSO_4$ contains $(32/233) \times M$ g of sulphur

$$\boxed{\% \text{ of sulphur in coal} = \frac{32}{233} \times \frac{M}{X} \times 100}$$

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Significance of Ultimate Analysis:

- Good-quality coal should have more percentage of C and H.
- Good-quality coal should have very little nitrogen content.
- Presence of sulphur is not desirable in coal
- A good-quality coal should contain lesser moisture content.

Carbonisation of Coal

Heating of coal in the absence of air at high temperature to produce a residue coke, tar and coal gas is called carbonisation.

(i) Caking of Coal: When coal is heated strongly, the mass becomes soft and coherent. Then it is called caking of coal.

(i) Coking of Coal: If the mass-produced is hard, porous, and strong then it is called the coking of coal.

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Destructive distillation of coal

Destructive distillation of coal is a chemical process that breaks down coal into simpler substances by heating it in the absence of air.

$$\text{Coal} \xrightarrow[\text{distillation}]{\text{destructive}} \text{Coke} + \text{Coal-tar} + \text{Coal-gas} + \text{Ammonical Solution}$$

Metallurgical coke

Metallurgical coke is a hard, porous, and strong carbon material used in blast furnaces and cupolas, to produce iron.

When bituminous coal is heated strongly in the absence of air, the volatile matter escapes out and a lustrous, dense, strong, porous and coherent mass is left, which is called metallurgical coke.

Most metallurgical coke is used in iron and steel industry processes

Thank You